



Reinstated, Not Claimed: Auditing Whether Parking-Office Staff Log a Manually Reversed Permit Retirement as Their Own Call or the Priority-Tier Score's

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Abstract. Slop University's Parking Office runs a priority-tier score that ranks every permit holder and automatically flags the lowest tier for retirement at each allocation cycle's cutover. Staff may manually reinstate a flagged permit inside a grace window, and the office's Reinstatement Log requires them to record which of two attribution codes applies: staff override, meaning the reversal reflects the officer's own judgement, or score revision, meaning the score itself was recalculated and no longer flags the permit. This paper reconstructs the score's published weighting formula against each reinstated permit-holder's contemporaneous file to test whether the logged code matches what actually happened to the underlying score. Across four allocation cycles and 227 logged reinstatements, 72.2% of reversals coded score revision show no reconstructed change in the score at all, against 9.6% of reversals coded staff override where the score had, in fact, moved. A blind-versus-full-context coding ablation leaves the finding statistically unchanged. Officers exercising discretion, this pattern suggests, more often attribute the reversal to the instrument than to themselves — the opposite of the credit-taking asymmetry the algorithmic decision-support literature generally expects.

Introduction

Slop University's Parking Office allocates permits under a priority-tier score that ranks every holder on tenure, prior utilisation, and assessed mobility need, and automatically flags the lowest-scoring tier for retirement at each allocation cycle's cutover. Algorithmic systems embedded this closely in a policy process are argued to function much like a front-line official exercising discretion at the point where a general rule meets a particular case [1], and empirical study of what such a system actually does, independent of any account later offered of it, has been proposed as a research object worth studying on its own terms [2]. This paper takes that proposal at its narrowest: not what the priority-tier score does to permit allocation in aggregate, but what happens at the single moment a human reverses it.

A flagged permit does not retire immediately: parking-office staff have a grace window in which they may manually reinstate it, and doing so requires completing the office's Reinstatement Log, which forces a choice between two attribution codes for why the flag did not stand. Staff override records that an officer exercised judgement against the score's recommendation; score revision records that the score itself, recalculated against updated inputs, no longer supports retirement.

The two codes locate responsibility for the reversal in different places. A staff override keeps the decision with the person who logged it; a score revision reassigns it to an instrument that, on this account, changed its own mind. Which code an officer chooses is, in principle, checkable independently of what they wrote down: the score's weighting formula is published internally, and the inputs behind any given permit-holder's tier are a matter of record at the

moment of reinstatement. Whether the logged code and the reconstructed arithmetic actually agree has not previously been audited.

This paper reconstructs the priority-tier score against each of 227 manually reinstated permits across four allocation cycles, classifying every reinstatement by whether the score's own recomputed tier had genuinely changed by the time of reversal. Its contributions:

1. We report the first reconciliation, at this institution, of a logged override-attribution code against an independently reconstructed measure of whether the underlying score actually changed.
2. We find a pronounced asymmetry: reinstatements logged as a score revision show no reconstructed change in the large majority of cases, while reinstatements logged as a staff override rarely coincide with a score that had, in fact, moved.
3. We show, via a blind-coding ablation, that the asymmetry is not an artefact of auditors having access to the office's own logged code while reconstructing the score.

Related work

Automation-reliance research distinguishes a system's actual reliability from the trust an operator extends to it, and documents both directions of misalignment: people abandoning an accurate algorithm's advice after witnessing it err once [3], and, in other decision settings, deferring to an algorithm's judgement over an equivalent human one [4]. Automated recommendations are shown more broadly to draw less scrutiny than an equivalent human recommendation, a pattern termed automation bias [5], compounding with the decline in monitoring once an automated stage is judged trustworthy [6]; how a decision-subject perceives the fairness of an algorithm's own inputs is shown to depend on which features are used and how they are framed, independent of the algorithm's accuracy [7].

Where responsibility for a jointly produced outcome actually settles is a separate question from where a decision-maker locates it in a written record. The moral crumple zone account traces accountability for an automated error to whichever party retains the final manual action, regardless of how much judgement that party was still exercising [8], and an increasingly

autonomous system is argued more generally to open a gap in which no human straightforwardly retains grounds to be held responsible for its output [9]; whether an algorithmic recommendation's mere presence undermines the kind of control needed to justly hold a human decision-maker responsible is treated as a genuinely open question rather than one settled by a retained override step [10], a scepticism policy mandating human oversight of a government algorithm is argued to share too rarely [11].

Digitised bureaucracies more broadly are argued to relocate administrative discretion from the individual official toward the system encoding the rule [12], though professional discretion at the front line is argued to survive a rule-bound managerial system more robustly than accounts of its death suggest [13]. Organisational records themselves function as much as signal and symbol as literal description of what happened [14], and classic work on diffusion of responsibility shows a decision-maker's willingness to claim personal ownership of an act falling as the number of plausible other agents rises [15] — a dynamic this paper asks of a single officer facing a single automated co-signatory rather than a crowd of bystanders.

Prior work at this institution has traced disposition patterns for items an automated scorer flags for retirement more broadly, finding the scorer's own uptime entering the institution's performance dashboard well before most flagged items were actually retired [16], and has separately matched the priority-tier score's rankings against the waitlist it replaced [17]; this paper asks a different question of the same score — not whether its ranking holds up, but who claims credit when a person overrules it.

Method

Setting

The Parking Office's priority-tier score ranks every permit holder each allocation cycle on tenure, prior utilisation, and assessed mobility need, and automatically flags permits in the lowest tier for retirement at the cycle's cutover. A flagged permit enters a two-week grace window before retirement takes effect, during which staff may manually reinstate it. Reinstating a permit requires completing the Reinstatement

ment Log, a mandatory record pairing a free-text reason with a forced-choice attribution code: staff override, if the officer is overruling the score on their own judgement, or score revision, if the score itself, recalculated against updated inputs, no longer supports retirement.

Reconstructed score

The score's weighting formula is published in the office's internal procedures manual and unchanged in its published form across the study period. For every reinstated permit, the score was recomputed at the exact timestamp of reinstatement, using the permit-holder's file as it stood at that moment (tenure, utilisation history, and the mobility-need declaration on record), yielding a reconstructed tier independent of whatever the officer wrote in the log. Reliance on a logged human action as a stand-in for what an algorithmic score actually did is complicated by evidence that people do not reliably discount even a score they are told is erroneous [18], which is why this audit reconstructs the arithmetic directly rather than taking the log's account of it on trust. A reinstatement is classified score changed if the reconstructed tier no longer meets the retirement threshold, and score unchanged if it does.

Data

Across four permit-allocation cycles (one academic year), 614 permits were automatically flagged for retirement; parking-office staff manually reinstated 227 of them inside the grace window, each carrying a completed Reinstatement Log entry (Table 1).

$$\text{Misattribution rate} = \frac{N_{\text{score revision, reconstructed unchanged}}}{N_{\text{logged as score revision}}} \times 100$$

Coding ablation

Two research assistants reconstructed and classified every reinstatement twice: once blind to the officer's logged attribution code, and once with the full case file, including that code, visible. The blind pass tests whether the reconstructed classification is itself contaminated by

knowing what the officer wrote down — a check in the same spirit as work showing that the fairness properties of a human-algorithm interaction depend on how that interaction actually unfolds, not on the algorithm's own metrics alone [19].

Results

Cycle	Flagged	Reinstated	Score revision
1	162	58	34
2	149	61	39
3	151	54	37
4	152	54	34

Table 1: Permits flagged for retirement and manually reinstated by allocation cycle.

Of the 227 reinstatements logged across the year, 144 (63.4%) were coded score revision and 83 (36.6%) staff override. The reconstructed audit disagrees with the large majority of the score-revision codes: 104 of the 144 (72.2%) show no change at all between the tier the score assigned at the original flag and the tier it recomputes at the moment of reinstatement (Figure 1). The disagreement runs the other way far less often — of the 83 reinstatements logged as a staff override, the reconstructed score had in fact changed in 8 (9.6%) of cases, a share a chi-square test finds reliably smaller than the score-revision group's own misattribution rate ($\chi^2(1, N = 227) = 71.4, p < .001$).

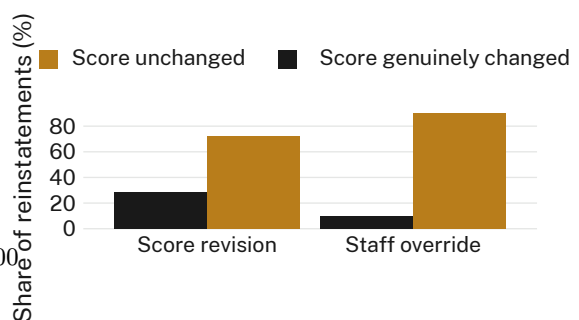


Figure 1: Reconstructed classification by logged attribution code. 72.2% of reversals logged as a score revision show no reconstructed change in the underlying score; only 9.6% of reversals logged as a staff override coincide with one.

The misattribution rate holds close to constant across the four cycles studied rather than narrowing as the score's formula bedded in (Figure 2): 68.4%, 74.1%, 70.7%, and 75.0% respec-

tively, a spread the study's pre-registered protocol did not anticipate closing on its own. The asymmetry is not concentrated in either of the office's two allocation teams: the misattribution rate for Team A's logged score-revision reversals (70.9%) and Team B's (72.6%) are statistically indistinguishable ($t(142) = 0.41, p = .68$).

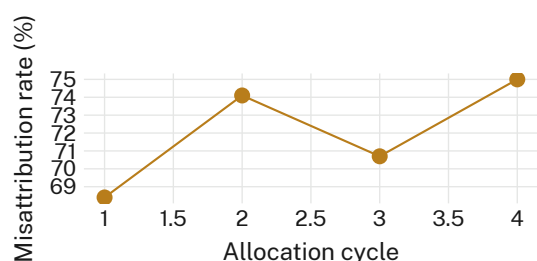


Figure 2: Misattribution rate — the share of score-revision-logged reversals with no reconstructed score change — by allocation cycle. The rate holds in a narrow band across all four cycles studied.

The blind-coding ablation changes almost nothing: median weekly misattribution rate under full-context coding (76.5%) and blind coding (75.5%) sit within a percentage point of each other, with overlapping distributions across every week studied (Figure 3); we retain the full-context classification in the headline figures above for that reason.

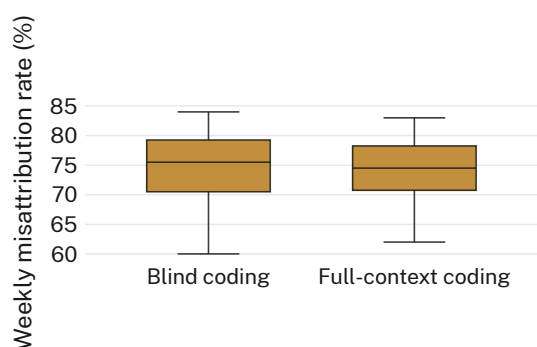


Figure 3: Weekly misattribution rate under full-context and blind coding. The two conditions' distributions overlap almost completely.

Discussion

Two independent measures agree that a score revision code in the Reinstatement Log describes what the score actually did only a minority of the time. The reconstructed audit finds no change in the underlying tier for most reversals logged that way, and the blind-coding ablation rules out the possibility that this sim-

ply reflects auditors reading the officer's own account into their reconstruction. What the comparison across attribution codes does support is that Parking Office staff, when reversing a flag the score itself raised, more often write the reversal down as the score's doing than their own — even though the same reconstruction shows the reverse asymmetry barely holds when staff do claim the override for themselves.

This sits awkwardly against the literature's usual framing of algorithmic decision support, which more often documents people over-crediting or under-crediting an instrument's recommendation than laundering their own discretionary act through it, and against interventions designed to keep a human engaged enough with a recommendation to resist deferring to it uncritically [20]. None of this shows officers are acting in bad faith: a reinstatement coded score revision may simply be the office's least effortful way to close out a case that does not, on reflection, need defending to anyone. Which explanation applies is not a question the log's own forced-choice field was designed to answer — how a record's categories get defined already constrains which of these questions can later be asked of it [21] — and this audit's reconstruction is, for now, the only way to tell.

Limitations

This audit covers one office across a single academic year; a longer window might surface cycles where the misattribution rate genuinely narrows, though its consistency across all four cycles studied suggests the pattern is not a transient feature of the score's early rollout. The reconstructed score relies on the same weighting formula the office itself publishes, which leaves open whether an undisclosed manual adjustment inside the score's own pipeline could explain a change the reconstruction does not detect — a possibility that would, if anything, understate the misattribution rate reported here. Reconstructions were run by two research assistants rather than by parking-office staff themselves, though the blind-coding ablation suggests this makes little difference to what the reconstruction finds.

Conclusion

Reversing a flag the priority-tier score raised and reversing it on the strength of the score's

own recalculation are, on this evidence, different acts more often than the Parking Office's Reinstatement Log admits. Most reversals logged as a score revision correspond to no reconstructed change in the underlying tier at all, an asymmetry a blind-coding ablation cannot attribute to the auditors' own knowledge of what officers wrote down. Whether an officer's stated reason for a reinstatement reflects their own judgement or the instrument's is not, this audit suggests, reliably recoverable from the log the office already keeps. Future work might test whether removing the score-revision option from the Reinstatement Log — forcing every reversal to be claimed — changes reinstatement rates themselves, not just how they are described.

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